

Decomposition-Transformation Assisted Optimized Heterogeneous Classification Strategy in NILM

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Abstract—Accurate identification of various loads for non-intrusive load monitoring (NILM) plays a crucial role in a variety of power system applications and strategies. In this article, a novel approach is proposed for performing load identification which consists of three main parts: features extraction considering decomposition-transformation strategy, appropriate feature screening, and employing a suitable classification strategy for performance improvement. The proposed approach first decomposes a load current into multiple modes by utilizing an adaptive signal decomposition technique, namely, variational mode decomposition (VMD), followed by the determination of the auto-correlograms of the modes. Multiple features are extracted from the obtained modes and their respective auto-correlograms. Postfeature extraction, combinations of multiple feature ranking (FR) techniques via union and intersection strategies are explored in contrast to employing only a single FR technique as prevalent in existing literature in NILM. Finally, considering the selected features by above stated FR combinations, a heterogeneous classifier strategy is adopted for performing classification that outperforms a single classifier as well as random forest classifiers on account of its heterogeneous structure. For further performance improvement, a surrogate-based optimization technique is utilized for determining the appropriate hyperparameter values in contrast to manual tuning which is prominent in existing works. Experimental appliance data obtained from publicly available datasets and execution via the software-in-loop mode have been considered to put forward the efficacy of the proposed approach.

Index Terms—Auto-correlation, heterogeneous classifiers, non-intrusive load monitoring (NILM), signal decomposition.

I. INTRODUCTION

RESIDENTIAL buildings have a significant share of energy consumption in both developed as well as developing nations. Utilizing energy consumption feedback, about 10%–30% of energy can be saved in residential environments [1] if appliance-level consumption (APC) information is available. APC information has also been reported to be utilized for energy disaggregation in smart homes [2], [3] and bulk power supply points [4], energy management and home

automation strategies [5], and so on. Utilizing the APC information, the concept of nonintrusive load monitoring (NILM) was first proposed in [6]. The NILM techniques reported in the existing literature, as shown in Fig. 1, are mainly grouped into three categories: supervised, unsupervised, and semisupervised techniques [7], [8]. Supervised techniques require labeled data of each individual appliance, while unsupervised techniques learn to distinguish between appliances without prior information, with labels applied manually or in accordance with predefined standards. Semisupervised learning techniques are trained on a small labeled dataset and a large unlabeled dataset. However, the performance of unsupervised and semisupervised techniques has been found to be less accurate than the supervised ones [7], [8], [9]. Furthermore, various public datasets, such as Plug-Level Appliance Identification Dataset (PLAID) [10] and Worldwide Household and Industry Transient Energy Dataset (WHITED) [11], comprising of the load signatures of a vast amount of appliances are available. These datasets comprise of generally four load (appliances) types, i.e., single state loads (e.g., bulb), multistate loads (e.g., washing machine), continuous loads (e.g., air conditioners), and nonlinear loads (e.g., laptops). Thus, due to the availability of public datasets comprising of variegated appliances as well as the low accuracy of semisupervised and unsupervised approaches, most of the recent works are devoted toward the supervised technique. In light of the recent literature [7], [8], [12], the present work also belongs to the supervised category. Within the supervised technique-based approaches, there are further two subcategories, namely, event-based [13] and event-less [14], [15] approaches. In line with the recent works [13], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], the approach presented in this article belongs to the class of event-based NILM methods for appliance recognition.

Recently, multiple approaches have been proposed in the literature which aims at improving the appliance classification strategy within the supervised event-based category. Within these approaches, steps have been actively taken to improve classification performance by focusing on either one or more of the following stages: feature extraction, feature ranking (FR), and performing classification via machine-learning (ML)-based techniques.

Initially, distinguishing features are extracted from the load signatures, namely, current, voltage, power, or a combination of these [16], [17], [18], [19], [20], [21]. After extracting the feature, the existing works have put forward their respective

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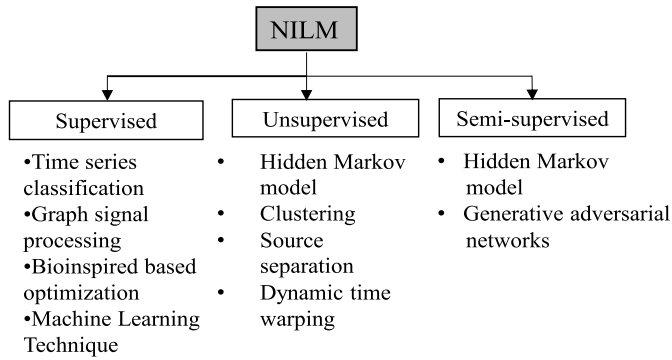


Fig. 1. Taxonomy of NILM techniques.

strategies for performing appliance identification [26], [27], [28], [29], [30]. In [26] and [27], features are directly extracted from the load signatures, followed by utilization of the classification algorithm. However, the reported accuracy is lower than most of the recent works. Recently, in order to enhance the feature set, the existing literature mentions different signal processing techniques such as fast Fourier transform (FFT) [23], Hilbert [25] transform, and statistical methods [31] to extract the features from the appliance current for classification [13], [24], [25]. The FR technique adopted in [26] and [28] is meant to earmark the most relevant features in order to improve the classifier's performance. Postfeature extraction, either a single feature selection method or feature fusion method is utilized. Finally, the final features are forwarded to a classifier.

Considering the state-of-the-art literature in appliance identification for NILM, there are certain aspects, wherein taking appropriate steps will lead to significant performance improvement. For example, existing approaches have generally explored basic signal processing techniques, such as FFT [23], Hilbert [25], etc., to process the load signatures and generate features. In this regard, it is imperative that advanced signal processing techniques be explored, which in turn assists in obtaining a rich set of features. Moreover, there is a multitude of feature selection methods available in the literature, and each of the selection methods operates differently from the others [26], [27], [28]. The available literature for appliance identification in NILM sheds no light regarding the consideration of more than one feature selection technique and their impact on classification accuracy. Next, the considered ranked feature set is utilized for performing appliance classification. Existing literature in appliance classification considers either one of the three ways while choosing classifiers: a single classifier [29], an arrangement of multiple homogeneous classifiers (random forest) [30], and an arrangement of a vast number of classifiers (stacked classifier) [32]. There are concerns with each of the approaches stated above. Either a single classifier or an arrangement of similar classifiers does not result in significant performance improvement and hence, an arrangement of heterogeneous classifiers is put forward in [32]. However, the approach in [32] has low classification accuracy the reason for which is the inappropriate hyperparameter settings of the classifiers [32]. The hyperparameters are considered from a limited number of manually selected values. Thus, it is

imperative that the hyperparameters are appropriately tuned for the problem at hand.

In this regard, this work proposes a novel strategy for appliance classification for NILM. Initially, an advanced signal decomposition technique, namely, VMD, is utilized for decomposing the recorded load signature into multiple modes. VMD has been utilized in recent studies in different fields, which include but are not limited to, cybersecurity [33], microgrid fault classification [34], etc. Apart from directly extracting features from the decomposed signals, this work proposes to construct auto-correlograms for each of the modes. This leads to a more enriched preliminary feature set, as the features are extracted from both signal decomposition (modes) as well as their further respective transformations (auto-correlograms). The extracted feature set, rather than being subjected to a single FR technique (as prevalent in existing works [26], [27], [28]), is subjected to multiple FR techniques in the present work, namely, relief-F [28], Fisher's score [35], and Pearson correlation coefficient (PCC) [36]. As stated previously, each of the feature selection techniques operates differently from the others, and thus, the features flagged by each of the FR techniques will also vary. To take advantage of the multiple feature selection methods considered, this work combines ranked features based on *union* and *intersection* strategies [37], [38], [39], [40], [41]. In other words, considering the strategy of either *union* or *intersection* signifies whether a feature has been earmarked by at least one or more FR techniques. To the best of our knowledge, none of the existing techniques for appliance recognition in NILM has utilized more than one FR method. A comparison of accuracy is presented while considering the features retained by individual the FR methods considered, and those retained by considering the abovementioned strategies. The final selected features are then forwarded for classification. In the present work, instead of a single classifier with manually selected hyperparameters, this work proposes to utilize a majority-voting-based hyperparameter optimized heterogeneous classifiers technique. For optimizing the hyperparameter settings, a surrogate function-based optimization technique, namely, Bayesian optimization, is considered comprising of the probability-of-improvement (PI)-based acquisition function (AF). The predictions provided by the heterogeneous classifiers are lastly processed by the majority voting criteria which in turn, further improves the classification accuracy. The main contributions of the proposed work are as follows.

- 1) In order to improve the classification accuracy, recent studies have explored various methods to enhance the feature sets [13], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25]. In the present work, we consider an advanced signal processing technique known as variational mode decomposition (VMD) to decompose the load signature into different modes followed by consideration of auto-correlation of the decomposed modes. Considering of such decomposition and transformation results in uniquely generated waveforms from the load signatures from which varied statistical features are extracted, leading to a rich feature set.

- 2) Different FR techniques operate uniquely and rank the features based on their own specified criteria. In contrast to the single FR technique as prevalent in NILM literature [26], [28], the present work explores union–intersection strategy regarding the utilization of a suitable combination of multiple FR techniques, i.e., features are earmarked by multiple FR techniques, which can further enhance the classification accuracy.
- 3) Different ML classifiers, function differently which is also apparent from the existing literature [7], [8]. In order to outweigh the incorrect predictions of particular classifiers, a majority voting-based combination of heterogeneous classifiers is proposed in the present work. Furthermore, hyperparameters of ML-based classifiers significantly affect the classification performance [13]. Rather than resorting to manually selecting the classifier's hyperparameters, the Bayesian optimization, with the probability of improvement as the AF, is used to intelligently tune the hyperparameters of the classifiers considered. Appropriate selection of the hyperparameters results in enhancing the classifier's performance, thus leading to improved classification accuracy.
- 4) The state-of-the-art approaches in NILM relevant to the present work has not reported running time as well as the platform utilized. To ensure the applicability of the proposed work for real-time applications, execution via the software-in-loop mode is also considered.

The rest of this article is organized as follows. Section II introduces the proposed method, including the event detection, obtaining modes via VMD and their transformations to auto-correlograms, various FR methods considered, strategies for combining ranked features, and the heterogeneous classification utilized for present work. The experimental results considering the publicly available datasets are presented in Sections III and IV concludes this article.

II. PROPOSED METHODOLOGY

In this section, the proposed approach for performing NILM is described.

A. Event Detection and Data Acquisition

Initially, the steady-state current data for each appliance need to be extracted prior to and post a switching event. In the present work, 0.1 s of steady-state data is utilized. As in other existing works, the present work considers that only one load is switched at a time. The variation in change in root mean square value is considered as the criterion for detecting switched events. A threshold is considered based on the public dataset utilized. After the detection of a switching event, the appliance currents are extracted from the steady-state signal, both before and after an event. The positive zero-crossing point of voltage data is considered as the beginning point for the preevent and postevent steady state appliance current, denoted here as i_{off} and i_{on} , respectively. The current of single appliance is thus defined as $i = i_{\text{on}} - i_{\text{off}}$ [24].

B. Appliance Current Features Extraction

Postevent detection, a signal processing technique is used for obtaining the features from extracted load signal.

It consists of VMD and obtaining modes, extracting multiple features from the modes and their respective auto-correlograms. In this section, various steps, namely, signal decomposition into intrinsic modes, auto-correlograms of respective modes, and lastly, feature extraction is presented.

1) *Signal Decomposition Into Intrinsic Modes*: The VMD is a quasi-orthogonal and adaptive signal decomposition technique, which is utilized to decompose a signal, $x(t)$, into finite modes and is not dependent on a predefined base function for decomposing the signal at hand. Briefly, the VMD is developed based on the notions of the Hilbert transform, the wiener filter, the analytic signal, and frequency shifting through harmonic mixing [42]. The VMD can decompose a signal $x(t)$ into finite modes as

$$x(t) = \sum_{k=1}^K q_k(t) \quad (1)$$

where K is the finite number of IMs desired and $q_k(t)$ denotes the principal IMs. Input $x(t)$ can be reconstructed by simply performing the addition of the estimated IMs. For each IM in VMD, the bandwidth is determined by considering the following steps: 1) determination of analytic signal for each mode q_k within the unilateral frequency spectrum by utilizing the Hilbert transform; 2) combination with a complex exponential of frequency ω_k in order to shift the mode's frequency spectrum to baseband; and 3) estimating the bandwidth by considering the squared L_2 norm of the gradient of the mixed signal. The consequent constrained variational optimization problem is

$$\begin{aligned} \min_{\{q_k\}, \{\omega_k\}} & \left\{ \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * q_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\} \\ \text{s.t.} & \sum_k q_k(t) = x(t) \end{aligned} \quad (2)$$

where $\{q_k(t)\}$ denotes the set of K IMs $\{q_1, \dots, q_K\}$, and $\{\omega_k\}$ denotes the set of central frequencies $\{\omega_1, \dots, \omega_K\}$ of IMs. ∂_t represents partial derivative with respect to time, and δ and $*$ denote the Dirac delta distribution and convolution, respectively. Also, $j^2 = -1$. For rendering the above-constrained optimization problem unconstrained, the inclusion of a quadratic penalty term and Lagrangian multiplier, λ , is suggested in [42]. The purpose of the quadratic penalty term is to enforce reconstruction fidelity, whereas the purpose of utilizing the Lagrangian multiplier is to enforce the constraints strictly. The unconstrained form of (2) is thus obtained from the augmented Lagrangian, $\mathcal{L}$, as [42]

$$\begin{aligned} \mathcal{L}(\{q_k\}, \{\omega_k\}, \lambda) = & \alpha \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * q_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \\ & + \left\| x(t) - \sum_k q_k(t) \right\|_2^2 \\ & + \left\langle \lambda(t), x(t) - \sum_k q_k(t) \right\rangle. \end{aligned} \quad (3)$$

The solution to (2) can now be found as the saddle point of the augmented Lagrangian function $\mathcal{L}$ in (3) by utilizing the alternate direction method of multipliers (ADMMs) approach.

Rather than optimizing $\mathcal{L}$ by considering all optimization variables simultaneously, the ADMM solves the problem by considering a sequence of iterative suboptimization problems, which in turn can be handled more easily. The minimization

Algorithm 1 ADMM Optimization Strategy for VMD

Initialize $\{q_k^1\}$, $\{\omega_k^1\}$, λ^1 , $n \leftarrow 0$

repeat

$n \leftarrow n + 1$

for $k = 1$ **to** K **do**

Update q_k :

$$q_k^{n+1} \rightarrow \arg \min_{q_k} \mathcal{L}(\{q_{i < k}^{n+1}\}, \{q_{i \geq k}^n\}, \{\omega_i^n\}, \lambda^n) \quad (4)$$

end for

for $k = 1$ **to** K **do**

Update ω_k :

$$\omega_k^{n+1} \rightarrow \arg \min_{\omega_k} \mathcal{L}(\{q_i^{n+1}\}, \{\omega_{i < k}^{n+1}\}, \{\omega_{i \geq k}^n\}, \lambda^n) \quad (5)$$

end for

Dual ascent:

$$\lambda^{n+1} \leftarrow \lambda^n + \tau \left(x - \sum_k q_k^{n+1} \right) \quad (6)$$

until convergence $\sum_k \|q_k^{n+1} - q_k^n\|_2^2 / \|q_k^n\|_2^2 < \eta$

problem mentioned in (4) is solved in the spectral domain (refer (19)–(22) in [42]) leading to

$$\hat{q}_k^{n+1} = \frac{\hat{x}(\omega) - \sum_{i \neq k} \hat{q}_i(\omega) + \frac{\hat{\lambda}(\omega)}{2}}{1 + 2\alpha(\omega - \omega_k)^2}. \quad (7)$$

Subsequently, by solving the suboptimization problem iteratively in (5), the center frequency ω_k can be updated in the dual frequency domain as

$$\omega_k^{n+1} = \left(\int_0^\infty \omega |\hat{q}_k(\omega)|^2 d\omega \right) / \left(\int_0^\infty |\hat{q}_k(\omega)|^2 d\omega \right). \quad (8)$$

The pseudocode for the ADMM strategy is shown in Algorithm 1. As stated above, the modes and their respective central frequencies are updated within the VMD algorithm. Interested readers are referred to [42] for further details. In the present work, VMD is used to decompose each of the appliance load signatures, i.e., i , into three IMs.

2) *Auto-Correlograms of Modes*: Prior to extracting features in the present work, an auto-correlation-based signal processing technique is also utilized. The auto-correlation technique has been recently used in the classification of bearing fault [43] and biomedical signals [44], respectively. Auto correlation technique finds out the degree of similarity of a signal with its delayed version itself. The auto-correlation

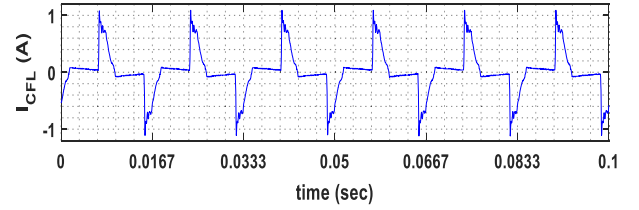


Fig. 2. Steady-state CFL current in PLAID [10].

of the appliance current, i , can be written as

$$\mathbf{z}_{ii}(m) = \begin{cases} \sum_{n=0}^{N-m-1} i_{n+m} i_n, & m \geq 0 \\ \mathbf{z}_{ii}(-m), & m < 0 \end{cases} \quad (9)$$

where $\mathbf{z}_{xx}$ is the auto-correlation sequence or auto-correlogram for signal i ; m signifies the time shifting factor or lag value. In a time series signal of N samples, it is observed that the resultant auto-correlation sequence is $(2N - 1)$.

3) *Feature Extraction*: From each of the decomposed modes and their respective auto-correlogram sequence, different set of features are extracted depending on the specific applications. The features considered in the present work are the following namely: basic statistical features (mean, standard deviation, root-mean-square, and shape factor), higher order statistical features (skewness and kurtosis), and impulsive metrics (peak value, impulse factor, crest factor, and clearance factor). Thus, in total, ten statistical features are extracted from each of the IMS as well as the auto-correlated signals of the IMs. In other words, for an appliance, say CFL, as shown in Fig. 2 (only 0.1-s data-samples shown) from the PLAID [10], six signals are obtained (three IMs and their respective auto-correlograms), as shown in Fig. 3. For each of the six signals shown in Fig. 3, the above-stated ten features are extracted. Thus, from each load signature, a total of 60 features are extracted. These features are forwarded to the FR methods mentioned next.

C. Multifeature Selection Methods

FR assists in determining the significance of the considered features. Unlike existing works [26], [27], [28], wherein a single FR technique is utilized for appliance identification in NILM, the proposed approach takes into account multiple FR techniques. Specifically, three ranking techniques, namely, relief-F [28], Fisher's score [35], and PCC [36] technique are utilized in the present work for ranking the features. The relief-F method aids in feature selection by spotting conditional dependencies between characteristics and uses a distance metric to weigh various features. In Fisher's score ranking technique, each of the features is chosen based on the ratings awarded by the Fisher criterion. The PCC method ranks the features as per the correlation values of the features. Once these three FR techniques are applied to the present feature set and features are distinctly ranked, the present work proceeds to take into account combinations of features ranked by at least one, two, or all the ranking techniques which are discussed next.

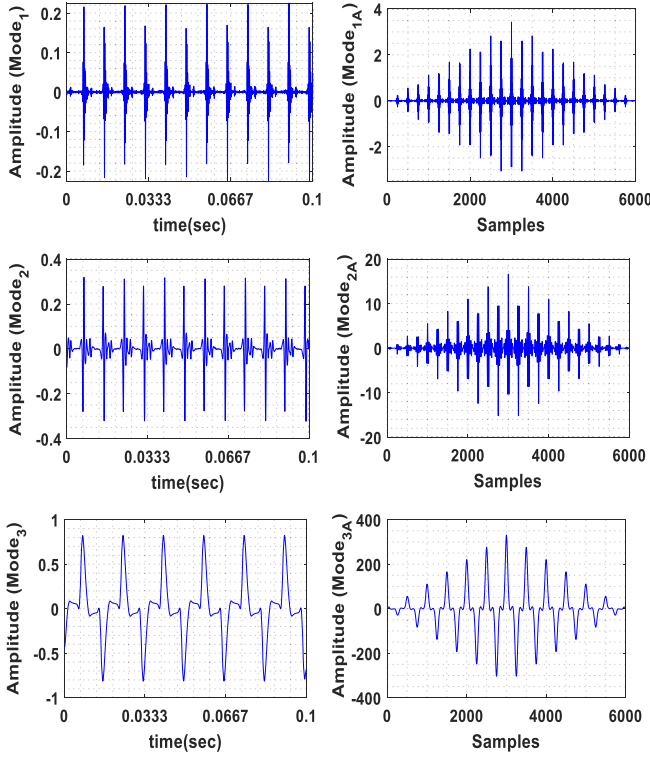


Fig. 3. Three intrinsic modes (Mode1–Mode3) obtained for CFL in PLAID [10] and their corresponding three auto-correlograms (Mode1A–Mode3A).

D. Combinations of Multiple FR Methods Using Union and Intersection Technique

As stated previously, the following union and intersection strategies are explored in the present work for taking advantage of multiple FR techniques considered together. Let the three FR methods (as mentioned in Section II-C) be denoted by α , β , and γ ; and there are a total of n features (in present work, $n = 60$) prior to ranking so that the feature set F is given as $F = \{f^1, f^2, \dots, f^n\}$. Also, let each of the FR techniques assigns scores (normalized) to the n features as per their respective criterion. The set of scores assigned by each FR technique to the n features are denoted as $R_\alpha = \{s_\alpha^1, s_\alpha^2, \dots, s_\alpha^n\}$, $R_\beta = \{s_\beta^1, s_\beta^2, \dots, s_\beta^n\}$, and $R_\gamma = \{s_\gamma^1, s_\gamma^2, \dots, s_\gamma^n\}$, where, s_α^i, s_β^i , and s_γ^i denote the normalized scores assigned to the i th feature of F (i.e., f^i) by the ranking methods α , β , and γ , respectively. Large normalized scores (s_α^i, s_β^i , and s_γ^i) correspond to the most important features, whereas lower values of the same correspond to the least important features. In any FR strategy, a threshold, say τ , is required to separate out the more relevant features from the less relevant ones. Now, the retained features post applying the ranking strategies can be considered as per one of the following conditions [37].

- 1) *Union of FR Methods*: A feature is assigned a normalized score above τ by at least one FR technique (i.e., either “ s_α^i ” or “ s_β^i ” or “ s_γ^i ” $> \tau$). This refers to any feature within $R_\alpha \cup R_\beta \cup R_\gamma$ whose normalized score is greater than τ . A set comprising of all the retained features as per this criterion is denoted from here on by $\Omega_1 \in F$.

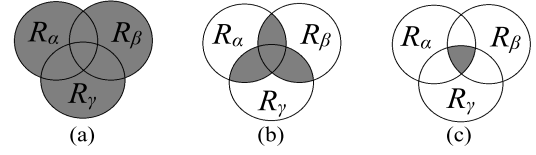


Fig. 4. Strategies for combining FR methods considered (a) $R_\alpha \cup R_\beta \cup R_\gamma$, (b) $R_\alpha \cap R_\beta$, $R_\beta \cap R_\gamma$, and $R_\alpha \cap R_\gamma$, and (c) $R_\alpha \cap R_\beta \cap R_\gamma$.

- 2) *Intersection of At Least Two FR Methods*: A feature is assigned a normalized score above τ by at least any two of the FR techniques considered (i.e., either “ s_α^i and s_β^i ” or “ s_β^i and s_γ^i ” or “ s_γ^i and s_α^i ” $> \tau$). This refers to any feature within $R_\alpha \cap R_\beta$, $R_\beta \cap R_\gamma$, $R_\alpha \cap R_\gamma$, and $R_\alpha \cap R_\beta \cap R_\gamma$ whose normalized score is greater than τ . A set comprising of all the retained features as per this criterion is denoted from here on by $\Omega_2 \in F$.
- 3) *Intersection of All FR Methods*: A feature is assigned a normalized score above τ by all three FR techniques considered (i.e., “ s_α^i, s_β^i and s_γ^i ” $> \tau$). This refers to any feature within $R_\alpha \cap R_\beta \cap R_\gamma$ only, whose normalized score is greater than τ . A set comprising of all the retained features as per this criterion is denoted from here on by $\Omega_3 \in F$.

The previously mentioned three strategies are shown schematically in Fig. 4. Based on the three strategies mentioned above and Fig. 4, the feature sets Ω_1 , Ω_2 , and Ω_3 are obtained. The comparison of the performance of each of these strategies is presented in Section III-C.

E. Heterogeneous Classifiers-Based Classification Strategy

The final set of features obtained by the union or intersection technique, as mentioned in the previous sub-section, is next utilized for performing appliance classification. In this regard, a majority voting-based hyperparameter-optimized heterogeneous classifier is proposed. In this section, the various aspects, namely, accuracy improvement due to majority voting, the classifiers considered, the surrogate approach for tuning of the hyperparameters, and lastly, the overall strategy, are presented.

1) *Accuracy Enhancement via Multiple Classifiers Considering Majority Voting*: Diverse classifiers result in diverse predictions. By combining the predictions of the heterogeneous classifiers by majority voting, improvement in classification accuracy can be obtained. The reason for this is explained as follows. Let a classifier prediction error be denoted by e and thus, the probability of the classifier providing correct prediction is $(1 - e)$. Considering multiple classifiers, the binomial distribution can be utilized to obtain the probability of the majority voting error. For example, using binomial distribution, the probability of p valid predictions out of v can be obtained in the following manner [45]:

$$\text{Probability of failure} = \binom{v}{p} e^p (1 - e)^{v-p} \quad (10)$$

where v denotes the total number of classifier/predictors, and p denotes the number of classifier/predictor which reports failure. The joint probability as per majority voting is obtained

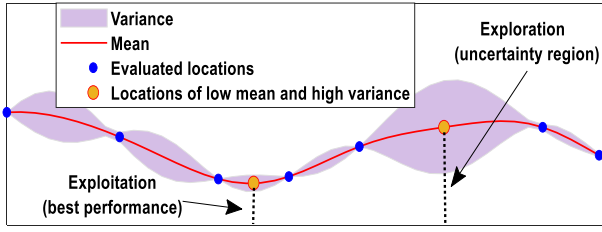


Fig. 5. Regions of exploration and exploitation of GP in the context of BO.

by summing up the probability for each value of p for every $p > v/2$

$$\text{Joint probability of failure } (\gamma) = \sum_{p > v/2}^v \binom{v}{p} e^p (1 - e)^{v-p}. \quad (11)$$

Thus, the net prediction error involving multiple classifiers, in principle, is diminished. However, it should be noted that for the above principle to be valid, the following two criteria must be satisfied: $e < 1/2$, and each of the classifiers operates independently. When the two criteria are satisfied, then in principle, $\gamma \rightarrow 0$ when $v \rightarrow \infty$. Hence, the overall classification accuracy improves compared to considering either a single classifier or an arrangement of homogeneous classifiers.

2) *ML Classifiers*: In the present work, four different benchmark classifiers, namely, support vector machines (SVMs), neural networks (NNs), decision trees (DTs), and k -nearest neighbors (kNNs) are considered. The predictions of each of these classifiers are forwarded to the majority voting algorithm for final prediction. The hyperparameter range considered for each of the classifiers is as follows: kernel scale = 1:100 and box-constraint = 1:100 for SVM, layer size = 1 – 200 and regularization strength = 0.01 – 1, the maximum number of splits = 1:500 for DT, and a number of neighbors = 1:500 and 11 distance metrics for kNN. The other specifications for the above-stated classifiers are linear kernel function for SVM, ReLU activation for NN, and Gini's diversity index for DT. The distance metrics considered for the kNN are Chebyshev, city block, Euclidean, correlation, cosine, Jaccard, Hamming, standardized Euclidean, Minkowski, Mahalanobis, and Spearman.

3) *Automatic Selection of Hyperparameters*: Selection of appropriate hyperparameters is crucial for the performance of the ML classifiers. It is to be noted that the ML classifiers are black box models and hence, a deterministic objective function cannot be formulated. Hence, in the present work, we utilize the Bayesian optimization with the Gaussian process (GP) as the surrogate function and the PI as the AF [46]. As the goal is to improve the classification accuracy, the objective f considered for the GP is the k -fold loss. Now, let the mean and the variance of the GP be denoted by μ and σ^2 , respectively, and θ and Θ be the hyperparameter settings and hyperparameter space, respectively. For $n + 1$ th iteration, the hyperparameter values are determined using the AF (Z) and the hyperparameter values at (n)th iteration as

$$\theta_{n+1} = \underset{\theta \in \Theta}{\operatorname{argmin}} Z(\theta_n). \quad (12)$$

The goal of utilizing the AF is to guide the search through large uncertainty regions (exploration) and best performance regions, i.e., low mean regions (exploitation) of the GP in the context of Bayesian optimization. For the purpose of understanding, the exploration and exploitation regions in the context of Bayesian optimization are shown in Fig. 5. Now, the probability of improvement for the AF is given by [46]

$$Z(\theta_n) = P(f(\theta_n) \geq f^* + \eta) \quad (13)$$

$$= \Psi\left(\frac{\mu(\theta_n) - f^* - \eta}{\sigma(\theta_n)}\right) \quad (14)$$

where $\Psi(\cdot)$ and η are the normal cumulative distribution function and a tradeoff parameter, respectively. f^* is the best observed value of $f(\theta)$, i.e., lowest observed value of posterior mean.

4) *Overall Classification Strategy*: The overall classification strategy considering all the aspects presented in the previous sub-sections results in the proposed majority voting-based hyperparameter optimized heterogeneous classifier as shown in Fig. 6.

III. RESULTS

A. Experimental Datasets Considered

For testing the efficacy of the proposed approach, the PLAID [10] and WHITED [11] datasets are considered in the present work. Both datasets consist of high-resolution appliance datasets which have been widely utilized for validating existing works on NILM [13], [20], [24]. The PLAID dataset consists of submetered measurements of 11 different appliances, recorded at a sampling rate of 30 kHz. The WHITED dataset contains data on appliances recorded at a sampling rate of 44.1 kHz. In the present work, we extract and utilize the current measurements available in the considered datasets mentioned above. For evaluating the performance of the proposed approach, the repeated tenfold cross-validation method is used [20]. In the tenfold cross-validation method, the actual dataset is partitioned randomly into ten subsets of equal size, out of which nine randomly chosen subsets are used for training and one subset is utilized for validation. For ascertaining a comprehensive evaluation of the experimental results, the following widely utilized metrics reported in the existing works are considered [24]: the F1-score, accuracy, and the Matthews correlation coefficient (MCC).

B. Results on Public Datasets

In contrast to utilizing a single FR technique, as prevalent in existing works on appliance identification [26], [27], [28], the present work uses multiple ranking methods, namely, relief-F, Fisher's score, and PCC. The normalized scores offered by each of the FR techniques are shown in Fig. 7. In the present work, the threshold, τ , is set to 0.6, as beyond this value, the performance of the overall approach deteriorates. Let the three feature vectors $R_{\alpha > \tau}$, $R_{\beta > \tau}$, and $R_{\gamma > \tau}$, contain only those features from the vectors R_α , R_β , and R_γ which have a normalized score greater than τ . All the feature vectors, which are determined as per the above strategy, are mentioned below.

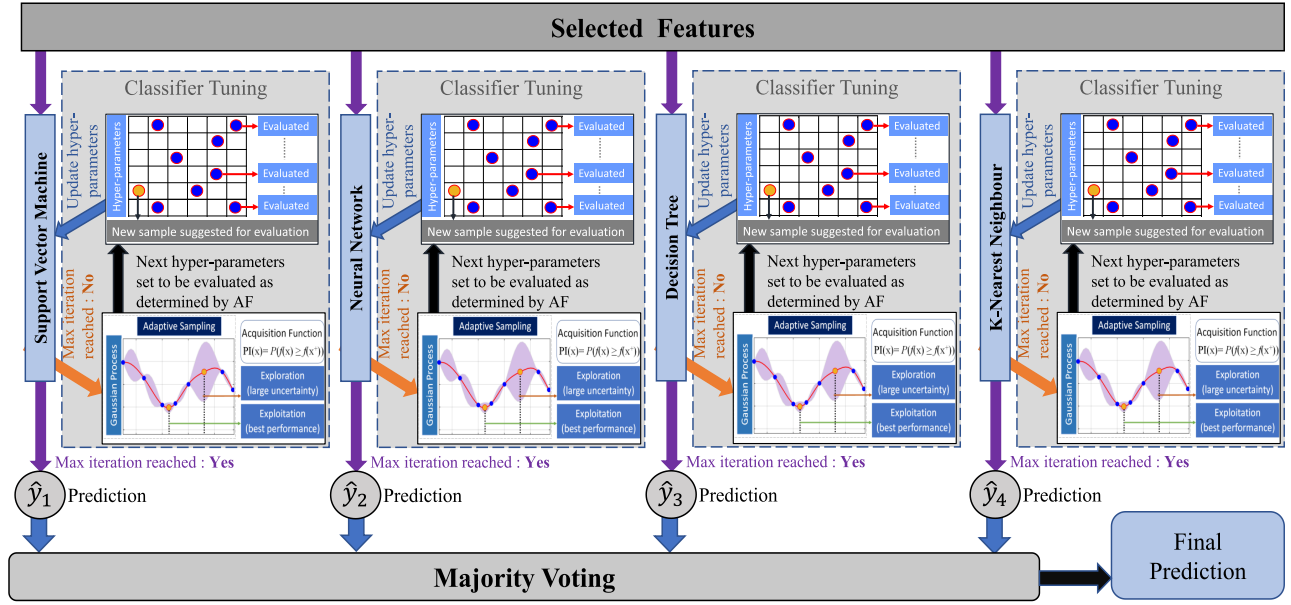


Fig. 6. Strategy for combining hyperparameter optimized heterogeneous classifiers.

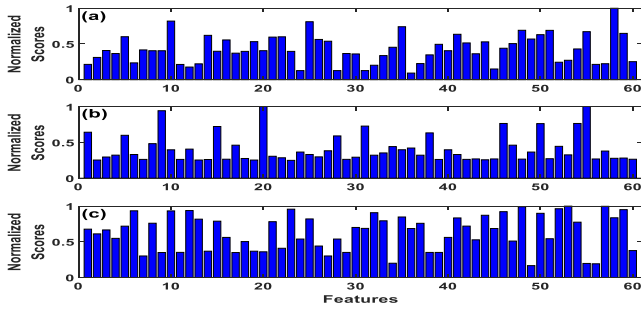


Fig. 7. FR methods utilized (a) relief-F, (b) Fisher score, (c) PCC.

- 1) Index of the features which have been provided normalized scores greater than τ by each of the FR methods

$$R_{\alpha > \tau} = [5; 10; 14; 21; 22; 25; 35; 41; 48; 49; 50; 51; 55; 58; 59]$$

$$R_{\beta > \tau} = [1; 5; 9; 15; 20; 31; 38; 46; 50; 54; 55]$$

$$R_{\gamma > \tau} = [1; 2; 3; 5; 6; 8; 10; 12; 13; 15; 21; 23; 25; 30; 31; 32; 33; 35; 36; 37; 41; 42; 44; 45; 46; 48; 50; 52; 53; 54; 57; 58; 59].$$

- 2) Index of the union of features having normalized scores greater than τ

$$\Omega_1 = [1 - 3; 5; 6; 8 - 10; 12 - 15; 20 - 23; 25; 30; 31 - 33; 35 - 38; 41; 42; 44 - 46; 48 - 51; 52 - 55; 57 - 59].$$

- 3) Index of the features that have been provided normalized scores greater than τ by at least two of the ranking methods, i.e., the intersection of features retained as per two ranking methods

$$\Omega_2 = [1; 5; 10; 15; 21; 25; 31; 35; 41; 46; 48; 50; 54; 55; 58; 59].$$

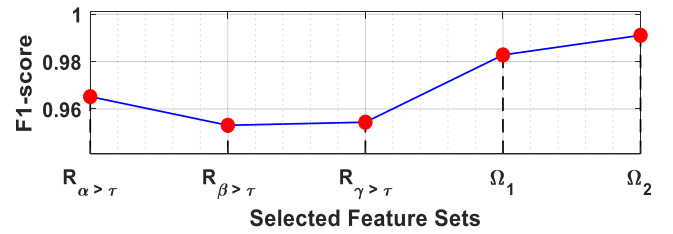


Fig. 8. Comparison of accuracies with different features sets on PLAID [10] datasets.

- 4) Index of the features that have been provided normalized scores greater than τ by all three of the ranking methods, i.e., the intersection of features retained as per three ranking methods

$$\Omega_3 = [5; 50].$$

Out of 60 features, $R_{\alpha > \tau}$ retains 15 features, $R_{\beta > \tau}$ retains 11 features, $R_{\gamma > \tau}$ retains 33 features, Ω_1 retains 41 features, Ω_2 retains 16 features, and Ω_3 retains only two features.

Considering each of the following feature sets: $R_{\alpha > \tau}$, $R_{\beta > \tau}$, $R_{\gamma > \tau}$, Ω_1 , Ω_2 , and Ω_3 , the classification using the heterogeneous classifier structure is performed. The hyperparameter and other settings of the considered heterogeneous classifier are described in Section II-E. The plot of the obtained accuracies are shown in Fig. 8. It has been observed that the best performance is obtained for the feature set Ω_2 , closely followed by the feature set Ω_1 . Also, the size of the set Ω_2 is significantly lesser than that of Ω_1 . Hence, the best feature set suggested by the present work is Ω_2 , i.e., all the features which have been earmarked by at least two of the FR methods. The performance of the feature sets $R_{\alpha > \tau}$, $R_{\beta > \tau}$, and $R_{\gamma > \tau}$ are lower than that of Ω_1 and Ω_2 . The worst performance was observed in Ω_3 . As the feature set Ω_2 provides the best performance, the same is finalized for appliance identification for NILM.

TABLE I

PERFORMANCE EVALUATION ON PLAID DATASET [10] WITH VMD AND POST AUTO-CORRELATION FEATURES WITH MANUAL TUNING AND HYPERPARAMETER OPTIMIZATION

Features considered (extracted from VMD modes)				Features considered (extracted post Auto-correlation)			
Manual Tuning		Hyperparameter Optimized		Manual Tuning		Hyperparameter Optimized	
MLC	F1-Score	MLC	F1-Score	MLC	F1-Score	MLC	F1-Score
SVM	94.10	SVM	94.26	SVM	94.12	SVM	94.48
NN	93.65	NN	94.03	NN	93.37	NN	94.21
kNN	94.32	kNN	94.78	kNN	95.10	kNN	95.85
DT	94.00	DT	94.17	DT	94.02	DT	94.32

TABLE II

PERFORMANCE EVALUATION ON PLAID DATASET [10] WITH RANKED, UNRANKED, MANUAL TUNING, AND HYPERPARAMETER OPTIMIZATION OF ALL FEATURES

Features considered (extracted from VMD modes + post Auto-Correlation)							
Without feature selection				With feature selection (Ω_2)			
Manual Tuning		Hyperparameter Optimized		Manual Tuning		Hyperparameter Optimized	
MLC	F1-Score	MLC	F1-Score	MLC	F1-Score	MLC	F1-Score
SVM	95.76	SVM	96.19	SVM	97.68	SVM	98.13
NN	95.66	NN	96.05	NN	97.26	NN	98.24
kNN	96.01	kNN	96.81	kNN	97.12	kNN	98.77
DT	95.03	DT	95.62	DT	96.48	DT	98.52
Final F1-Score by Combining all heterogeneous classifiers (hyper parameters optimized) prediction using majority voting: 99.11							

C. Comprehensive Performance Analysis

In the present work, the selection of hyperparameters of each classifier, the maximum number of iterations is set to 50 for Bayesian optimization. hence, within the 50 iterations, the appropriate hyperparameter settings are determined. Table I shows the stepwise improvements of different steps considered for the proposed hyperparameter optimized NILM technique. In Table I, initially, only features extracted from decomposed of VMD modes are considered without using any feature selection techniques. It is observed that the accuracy obtained in hyperparameter-optimized classifiers is higher compared to the accuracy obtained by manually tuned classifiers. Similarly, when features are extracted using auto-correlation on each decomposed mode obtained from VMD signals, the accuracy obtained in hyperparameter-optimized classifiers is higher compared with the accuracy obtained by manually tuned classifiers.

Next, feature extraction from both techniques is considered, i.e., decomposed modes of each VMD signal (modes) and auto-correlation of each decomposed mode obtained from VMD. The obtained accuracies from all the classifiers are shown in Table II. It was observed from Table II that without using any feature selection, the accuracies obtained in hyperparameter optimized classifiers are higher compared to the accuracies obtained by manually tuned classifiers. It was also observed from Table II that accuracies improved significantly when features are extracted from both VMD and auto-correlation, compared to Table I. Next, only those features from each of the sets (VMD and auto-correlation) in the final features set (Ω_2) is used for different machine-learning classifiers for classification. From Table II, it is concluded

TABLE III

PERFORMANCE EVALUATION ON PLAID DATASET AT 30 kHz

Appliance	Prec. (%)	Rec. (%)	Acc (%)	F1 (%)	MCC (%)
AC	99.05	99.03	99.52	98.90	98.90
CFL	99.09	99.54	99.81	99.32	99.22
Fan	99.56	98.11	99.61	98.82	98.66
Laptop	99.05	99.03	99.67	99.03	98.90
Fridge	98.89	98.88	99.21	98.82	98.82
Hairdryer	99.60	99.20	99.65	99.40	99.29
ILB	98.65	99.32	99.57	98.98	98.88
Microwave	99.56	100	99.62	99.68	99.74
Heater	98.82	98.81	99.78	98.76	98.76
VC	98.64	100	99.59	99.22	99.29
WM	98.67	98.67	99.68	98.60	98.60
Average	99.06	99.15	99.61	99.11	98.91

that hyperparameter-optimized classifiers consistently provide better accuracy than manually tuned classifiers. Finally, the majority voting technique is used to combine all heterogeneous classifiers' accuracies, which significantly improved the overall performance. The complete set of appliance classification accuracy metrics for Ω_2 and the optimized heterogeneous classifier, is reported in Table III for the PLAID dataset. The accuracy metrics consist of precision (Prec.), recall (Rec.), accuracy (Acc.), F1 score, and MCC. The average classification accuracy and F1-score obtained on the PLAID dataset are 99.61% and 99.11%, respectively. Similarly, for WHITED dataset also, the corresponding feature vector Ω_2 is found to have the best performance. The average classification accuracy and F1-score obtained are 99.89% and 99.68%, respectively. MCC produces a more informative score in evaluating classifications task as compared to the accuracy and F1 score [24]. For the PLAID and WHITED datasets, obtained MCC scores are 0.9891 and 0.9942, respectively.

D. Validation Using Software-in-Loop Mode

The training of the ML classifiers and determination of the associated hyperparameter via the surrogate-based optimization technique is performed offline by using ranked features. To validate the performance of the proposed work, a software-in-loop mode has been utilized. The reason for considering the software-in-loop mode is to check the performance and execution times of the proposed work when converted to a low-level programming language such as C/C++. Such validation is crucial as the embedded devices are capable of executing low-level programming language only. Also, it is to be noted that this mode is meant for checking the performance of the proposed overall approach (consisting of the tuned ML classifier which is performed offline) for appliance classification. Hence, the associated execution time is the time when an event is detected to the time when the appliance is classified via the trained heterogeneous classifiers. The execution time of appliance prediction is reported here which is performed in two different ways: 1) identification performed via the MATLAB codes and 2) identification performed via C codes (converted from the MATLAB codes) which are executed in the host computer (known as software-in-loop validation). The host computer specification is 12th Gen Intel(R) i7-1255U, 24-GB RAM, with 512 GB (SSD) memory unit. It is found

TABLE IV

F1-SCORE COMPARISON OF SOME APPLIANCES FROM PLAID DATASET AS REPORTED IN THE STATE-OF-THE-ART TECHNIQUES

Ref.	Fridge	WM	Heater	AC
[26]	54.84	70.07	76.00	58.64
[19]	50.93	68.82	82.23	46.65
[2]	92.70	80.00	100.00	90.00
[3]	95.24	95.24	100.00	90.48
[20]	79.41	96.00	100.00	87.22
[18]	91.00	99.00	95.00	96.00
[23]	92.09	98.30	98.31	94.68
[13]	97.68	98.11	98.67	97.96
[24]	88.62	94.03	95.94	93.75
[31]	93.80	97.72	97.95	96.30
Proposed	98.82	98.60	98.76	98.90

that for both 1) and 2), the appliances are correctly identified with the only difference being in the execution time. The average execution time taken for 1) is 1.12 s whereas, for 2), it is 0.83 s.

E. Comparison With the Existing Literature

The classification performance of the proposed work is compared with that of other methods [2], [3], [13], [18], [19], [20], [23], [24], [26], [31], in NILM using public datasets such as PLAID and WHITED. To ensure a fair comparison, the appliances selected in Table IV are fridge, washing machine, heater, and air conditioner, which are available in both the PLAID dataset and the existing literature. The conclusion drawn from Table IV is that the proposed work improves the identification accuracy of appliances such as the fridge, washing machine, heater, and air conditioner compared to the existing literature in NILM for public datasets such as PLAID. Although better identification results have been obtained for the heater using [2], [3], [20], lower accuracy is observed for the identification of appliances such as the fridge, washing machine, and air conditioner, as well as their overall performance for all types of loads. Table IV also shows that previous works [13], and [18] achieved the best performance for the fridge, washing machine, and air conditioner within the PLAID dataset. Compared to [13], the proposed work achieves better identification accuracy for the fridge and air conditioner, while the identification accuracy for the washing machine is higher in [18]. Therefore, [2], [3], [20] is better suited for loads that have two states, namely, ON and OFF, while the proposed technique provides significantly better results for identifying all types of loads.

Table V describe the complete comparison with recent existing works in terms of overall F1 score. Compared to the existing literature [2], [3], [13], [18], [19], [20], [23], [24], [26], [31] the heater has highest identification accuracy achieved in [2], [3], and [20] shows in Table IV, but the proposed technique outperforms the methods in [2], [3], and [20] in terms of average F1-score with improved by 8.21%, 6.63%, and 3.71% across the PLAID datasets, respectively. Compared to the existing literature [2], [3], [13], [18], [19], [20], [23], [24], [26], [31], the highest accuracy is achieved for washing machines in [18], but average identification for all appliances in PLAID datasets is lower by 1.75% compared

TABLE V

COMPARISON OF CLASSIFICATION PERFORMANCE OF EXISTING TECHNIQUES WITH PROPOSED WORK

Ref.	Design technique	F1-score(%)	
		PLAID	WHITED
[19]	VI trajectories image	77.60	75.46
[26]	Time series features	78.71	-
[2]	Graph and group decision classifier	90.90	-
[3]	FFT-BDT	92.48	-
[20]	Colour encoded V-I Image	95.40	98.66
[18]	Reconstructed V-I Image	97.36	-
[23]	Current harmonics features	97.37	-
[13]	WRG V-I image	97.77	-
[24]	Temporal & spectral Features	97.79	99.48
[31]	Statistical features	98.19	-
	Proposed	99.11	99.68

TABLE VI

COMPARISON OF THE PROPOSED APPROACH WITH AN EXISTING WORK [25]

Sl. no	Precision (%)		Recall (%)		Acc (%)		F1score (%)	
	[25]	Prop.	[25]	Prop.	[25]	Prop.	[25]	Prop.
1	98.61	99.04	92.39	97.63	91.21	99.61	95.40	98.33
2	91.81	98.18	97.60	98.63	89.78	99.60	94.61	98.41
3	97.89	98.10	93.09	97.63	91.27	99.51	95.43	97.86
4	92.18	98.07	88.49	99.51	82.31	99.67	90.30	98.78
5	97.46	96.67	95.90	96.66	93.55	98.67	96.67	96.67
6	97.85	98.39	91.29	99.91	89.45	99.78	94.46	99.19S
7	92.71	98.65	98.00	96.69	90.99	99.51	95.28	97.66
8	92.67	98.25	98.70	99.56	91.55	99.72	95.59	98.90
9	97.58	99.95	97.00	94.44	94.72	99.72	97.29	91.14
10	93.56	98.63	97.49	98.63	91.36	99.89	95.49	98.63
11	93.09	96.00	94.39	97.30	88.12	99.72	93.74	96.64

The appliance index are as follows: 1. AC, 2. CFL, 3. Fan, 4. Laptop, 5. Fridge, 6. Hairdryer, 7. ILB, 8. Microwave, 9. Heater, 10. Vacuum cleaner, 11. Washing machine

to the proposed technique. For PLAID datasets, [31] has the best performance among the previous works. The proposed work outperforms [31] by an average increment in accuracy of 0.92%. Similarly, for WHITED, [24] has the best performance in previous works. The proposed work outperforms [24] by an average increment in the accuracy of 0.34% for all appliances. Out of the several previous works mentioned in Table V, the MCC score is only available in [13] and [24]. For the PLAID dataset, the MCC score obtained via the proposed approach is 0.9891 against that of 0.9800 in [13] and 0.9810 in [24]. Therefore, the proposed method achieves better performance compared to state-of-the-art techniques in terms of accuracy or F1-score on public datasets.

Performing appliance classification by down-sampling public datasets is recently reported in [25]. The public datasets are down-sampled to 2 kHz and the down-sampled datasets are utilized for performing classification using HT-LSTM [25]. Table VI shows the comparison of the proposed approach and [25] considering the PLAID [10] dataset down-sampled to 2 kHz. Table VI shows the detailed comparison of precision, recall, accuracy, and F1-score of individual appliances, respectively, for the down-sampled PLAID dataset. The performance classification results of laptop load are improved significantly by 8.48%. In addition, CFL, washing machine, and hairdryer

load identification accuracy is improved by 3.8%, 2.9%, and 4.73% compared to [25]. Only the heater load obtained identification accuracy is slightly lower by 0.15% compared to [25]. The average F1 score of the proposed method on the same dataset obtained is 97.80% and the average F1 score of [25] is 94.93%. Thus, there is an improvement of 2.87% for the F1-score. Therefore, the improvement in average accuracy and F1-score measurement shows that the proposed load classification method has a better performance compared with the existing ones.

IV. CONCLUSION

This article puts forward a novel framework for performing appliance identification for NILM. Specifically, this article proposes to bring together several techniques which remain unexplored in the field of NILM, such as feature extraction from signal decomposition followed by transformation, union-intersection FR combinations, and stacked ensemble strategy for performing appliance identification for NILM. For appliance current decomposition, an adaptive signal decomposition technique, namely, VMD, has been employed, which provides multiple modes, which in turn, are further utilized for obtaining the auto-correlograms of the respective modes. A large set of features are extracted from the modes and their auto-correlograms. For performing feature selection, multiple FR methods instead of one, as prevalent in existing works in NILM, have been employed by taking into consideration the union-intersection strategies. Utilizing the final selected features, a heterogeneous classification strategy is employed which consists of multiple distinct classifiers and a majority voting. The hyperparameters of each classifier are intelligently determined by the Bayesian optimization technique with PI as the AF. The associated execution time for appliance identification is determined in two ways, namely, execution of MATLAB codes and software-in-loop mode in the host computer. The proposed framework results in improved performance to the recent works for appliance identification in NILM.

REFERENCES

- [1] D. He, "An advanced non-intrusive load monitoring technique and its application in smart grid building energy management systems," Ph.D. dissertation, Georgia Inst. Technol., Atlanta, GA, USA, 2016. [Online]. Available: <https://smartech.gatech.edu/handle/1853/54951>
- [2] X. Wu, Y. Guo, M. Yan, X. Li, L. Yao, and G. Gong, "Non-intrusive load monitoring using identity library based on structured feature graph and group decision classifier," *IEEE Trans. Smart Grid*, vol. 14, no. 3, pp. 1958–1973, May 2023.
- [3] T. Le, H. Kang, and H. Kim, "Household appliance classification using lower odd-numbered harmonics and the bagging decision tree," *IEEE Access*, vol. 8, pp. 55937–55952, 2020.
- [4] Y. Xu and J. V. Milanovic, "Artificial-intelligence-based methodology for load disaggregation at bulk supply point," *IEEE Trans. Power Syst.*, vol. 30, no. 2, pp. 795–803, Mar. 2015.
- [5] M. A. Mengistu, A. A. Girmay, C. Camarda, A. Acquaviva, and E. Patti, "A cloud-based on-line disaggregation algorithm for home appliance loads," *IEEE Trans. Smart Grid*, vol. 10, no. 3, pp. 3430–3439, May 2019.
- [6] G. W. Hart, "Nonintrusive appliance load monitoring," *Proc. IEEE*, vol. 80, no. 12, pp. 1870–1891, Dec. 1992.
- [7] P. A. Schirmer and I. Mporas, "Non-intrusive load monitoring: A review," *IEEE Trans. Smart Grid*, vol. 14, no. 1, pp. 769–784, Jan. 2023.
- [8] S. Dash and N. C. Sahoo, "Electric energy disaggregation via non-intrusive load monitoring: A state-of-the-art systematic review," *Electr. Power Syst. Res.*, vol. 213, Dec. 2022, Art. no. 108673.
- [9] A. Iwayemi and C. Zhou, "SARAA: Semi-supervised learning for automated residential appliance annotation," *IEEE Trans. Smart Grid*, vol. 8, no. 2, pp. 779–786, Mar. 2017.
- [10] J. Gao, S. Giri, E. C. Kara, and M. Bergés, "PLAID: A public dataset of high-resolution electrical appliance measurements for load identification research: Demo abstract," in *Proc. 1st ACM Conf. Embedded Syst. Energy-Efficient Buildings*, New York, NY, USA, Nov. 2014, pp. 198–199, doi: [10.1145/2674061.2675032](https://doi.org/10.1145/2674061.2675032).
- [11] M. Kahl, A. U. Haq, T. Kriechbaumer, and H.-A. Jacobsen, "Whited—A worldwide household and industry transient energy data set," in *Proc. 3rd Int. Workshop Intrusive Load Monitor.*, 2016, pp. 1–4.
- [12] S. Ghosh, A. Chatterjee, and D. Chatterjee, "Improved non-intrusive identification technique of electrical appliances for a smart residential system," *IET Gener., Transmiss. Distrib.*, vol. 13, no. 5, pp. 695–702, Mar. 2019.
- [13] A. Faustine, L. Pereira, and C. Klemenjak, "Adaptive weighted recurrence graphs for appliance recognition in non-intrusive load monitoring," *IEEE Trans. Smart Grid*, vol. 12, no. 1, pp. 398–406, Jan. 2021.
- [14] S. Makonin, F. Popowich, I. V. Bajić, B. Gill, and L. Bartram, "Exploiting HMM sparsity to perform online real-time nonintrusive load monitoring," *IEEE Trans. Smart Grid*, vol. 7, no. 6, pp. 2575–2585, Nov. 2016.
- [15] A. Cominola, M. Giuliani, D. Piga, A. Castelletti, and A. E. Rizzoli, "A hybrid signature-based iterative disaggregation algorithm for non-intrusive load monitoring," *Appl. Energy*, vol. 185, pp. 331–344, Jan. 2017.
- [16] L. d. S. Nolasco, A. E. Lazzaretti, and B. M. Mulinari, "DeepDFML-NILM: A new CNN-based architecture for detection, feature extraction and multi-label classification in NILM signals," *IEEE Sensors J.*, vol. 22, no. 1, pp. 501–509, Jan. 2022.
- [17] E. Azizi, M. T. H. Beheshti, and S. Bolouki, "Appliance-level anomaly detection in nonintrusive load monitoring via power consumption-based feature analysis," *IEEE Trans. Consum. Electron.*, vol. 67, no. 4, pp. 363–371, Nov. 2021.
- [18] D. Jia, Y. Li, Z. Du, J. Xu, and B. Yin, "Non-intrusive load identification using reconstructed voltage-current images," *IEEE Access*, vol. 9, pp. 77349–77358, 2021.
- [19] L. De Baets, J. Ruysinck, C. Devellder, T. Dhaene, and D. Deschrijver, "Appliance classification using VI trajectories and convolutional neural networks," *Energy Buildings*, vol. 158, pp. 32–36, Jan. 2018.
- [20] Y. Liu, X. Wang, and W. You, "Non-intrusive load monitoring by voltage-current trajectory enabled transfer learning," *IEEE Trans. Smart Grid*, vol. 10, no. 5, pp. 5609–5619, Sep. 2019.
- [21] S. Ghosh, D. Manna, A. Chatterjee, and D. Chatterjee, "Remote appliance load monitoring and identification in a modern residential system with smart meter data," *IEEE Sensors J.*, vol. 21, no. 4, pp. 5082–5090, Feb. 2021.
- [22] V. Singhal, J. Maggu, and A. Majumdar, "Simultaneous detection of multiple appliances from smart-meter measurements via multi-label consistent deep dictionary learning and deep transform learning," *IEEE Trans. Smart Grid*, vol. 10, no. 3, pp. 2969–2978, May 2019.
- [23] S. Ghosh, A. Chatterjee, and D. Chatterjee, "An improved load feature extraction technique for smart homes using fuzzy-based NILM," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–9, 2021.
- [24] J. Chen, X. Wang, X. Zhang, and W. Zhang, "Temporal and spectral feature learning with two-stream convolutional neural networks for appliance recognition in NILM," *IEEE Trans. Smart Grid*, vol. 13, no. 1, pp. 762–772, Jan. 2022.
- [25] T.-T.-T. Le, S. Heo, and H. Kim, "Toward load identification based on the Hilbert transform and sequence to sequence long short-term memory," *IEEE Trans. Smart Grid*, vol. 12, no. 4, pp. 3252–3264, Jul. 2021.
- [26] H. Liu, H. Wu, and C. Yu, "A hybrid model for appliance classification based on time series features," *Energy Buildings*, vol. 196, pp. 112–123, Aug. 2019.
- [27] N. Sadeghianpourhamami, J. Ruysinck, D. Deschrijver, T. Dhaene, and C. Devellder, "Comprehensive feature selection for appliance classification in NILM," *Energy Buildings*, vol. 151, pp. 98–106, Sep. 2017.
- [28] S. Bao et al., "Feature selection method for nonintrusive load monitoring with balanced redundancy and relevancy," *IEEE Trans. Ind. Appl.*, vol. 58, no. 1, pp. 163–172, Jan. 2022.

- [29] Y. Himeur, A. Alsalemi, F. Bensaali, and A. Amira, "Robust event-based non-intrusive appliance recognition using multi-scale wavelet packet tree and ensemble bagging tree," *Appl. Energy*, vol. 267, Jun. 2020, Art. no. 114877.
- [30] P. R. Z. Taveira, C. H. V. De Moraes, and G. Lambert-Torres, "Non-intrusive identification of loads by random forest and fireworks optimization," *IEEE Access*, vol. 8, pp. 75060–75072, 2020.
- [31] S. Ghosh, A. Chatterjee, and D. Chatterjee, "Extraction of statistical features for type-2 fuzzy NILM with IoT enabled control in a smart home," *Expert Syst. Appl.*, vol. 212, Feb. 2023, Art. no. 118750.
- [32] Y. Li, H. Wang, Z. Yang, J. Yang, and Z. Chen, "Stacking ensemble learning-based load identification considering feature fusion by cyber-physical approach," *IEEE Sensors J.*, vol. 23, no. 6, pp. 5997–6007, Mar. 2023.
- [33] W. Qiu et al., "Cyber-attack identification of synchrophasor data via VMD and multifusion SVM," *IEEE Trans. Ind. Appl.*, vol. 58, no. 2, pp. 1456–1465, Mar. 2022.
- [34] R. Azizi and S. Seker, "Microgrid fault detection and classification based on the boosting ensemble method with the Hilbert–Huang transform," *IEEE Trans. Power Del.*, vol. 37, no. 3, pp. 2289–2300, Jun. 2022.
- [35] J. J. Saucedo-Dorantes, M. Delgado-Prieto, R. A. Osornio-Rios, and R. de Jesus Romero-Troncoso, "Multifault diagnosis method applied to an electric machine based on high-dimensional feature reduction," *IEEE Trans. Ind. Appl.*, vol. 53, no. 3, pp. 3086–3097, May 2017.
- [36] M. Kordestani, A. Chaibakhsh, and M. Saif, "SMS—A security management system for steam turbines using a multisensor array," *IEEE Syst. J.*, vol. 14, no. 3, pp. 3813–3824, Sep. 2020.
- [37] C.-F. Tsai and Y.-C. Hsiao, "Combining multiple feature selection methods for stock prediction: Union, intersection, and multi-intersection approaches," *Decis. Support Syst.*, vol. 50, no. 1, pp. 258–269, Dec. 2010.
- [38] C.-F. Tsai and Y.-T. Sung, "Ensemble feature selection in high dimension, low sample size datasets: Parallel and serial combination approaches," *Knowl.-Based Syst.*, vol. 203, Sep. 2020, Art. no. 106097.
- [39] G. Yao, X. Hu, and G. Wang, "A novel ensemble feature selection method by integrating multiple ranking information combined with an SVM ensemble model for enterprise credit risk prediction in the supply chain," *Expert Syst. Appl.*, vol. 200, Aug. 2022, Art. no. 117002.
- [40] C. Chen, Y. Tsai, F. Chang, and W. Lin, "Ensemble feature selection in medical datasets: Combining filter, wrapper, and embedded feature selection results," *Expert Syst.*, vol. 37, no. 5, Oct. 2020, e12553.
- [41] S. Salesi and G. Cosma, "Generalisation power analysis for finding a stable set of features using evolutionary algorithms for feature selection," *Knowl.-Based Syst.*, vol. 231, Nov. 2021, Art. no. 107450.
- [42] K. Dragomiretskiy and D. Zosso, "Variational mode decomposition," *IEEE Trans. Signal Process.*, vol. 62, no. 3, pp. 531–544, Feb. 2014.
- [43] S. S. Roy, S. Dey, and S. Chatterjee, "Autocorrelation aided random forest classifier-based bearing fault detection framework," *IEEE Sensors J.*, vol. 20, no. 18, pp. 10792–10800, Sep. 2020.
- [44] G. Zhou, J. H. L. Hansen, and J. F. Kaiser, "Nonlinear feature based classification of speech under stress," *IEEE Trans. Speech Audio Process.*, vol. 9, no. 3, pp. 201–216, Mar. 2001.
- [45] C. C. Aggarwal, *Data Classification: Algorithms and Applications*. Boca Raton, FL, USA: CRC Press, 2015.
- [46] E. Brochu, V. M. Cora, and N. de Freitas, "A tutorial on Bayesian optimization of expensive cost functions, with application to active user modeling and hierarchical reinforcement learning," 2010, *arXiv:1012.2599*.



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